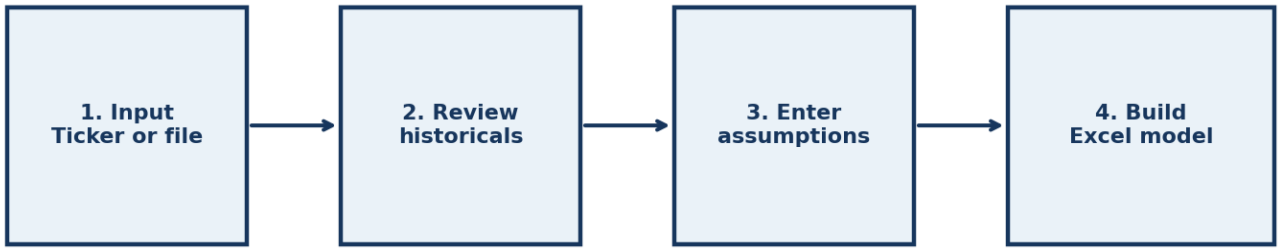
Generic DCF Automation

Beginner-friendly User Guide

What this tool does

The program creates an Excel DCF model from scratch. You can provide a public-company ticker or upload historical financials. The program then asks for valuation assumptions and creates a workbook with financials, assumptions, forecast, DCF sensitivity and charts.



1. Install Python and VS Code

Install Python 3.10 or later and Visual Studio Code. In VS Code, open a folder where you want to keep the project. Download the Python script, requirements file and sample CSV into that folder.

2. Install the required packages

Open Terminal → New Terminal in VS Code and run:

```
python -m pip install -r requirements_generic_dcf.txt
```

If the command fails, check that Python is installed and that the terminal is open in the folder containing the requirements file.

3. Choose your input method

Option A — ticker mode. For a US ticker, use a symbol such as MSFT. For an Indian listing, use the Yahoo Finance symbol where available, such as RELIANCE.NS. Data availability varies by company and region.

```
python generic_dcf_from_scratch.py --ticker MSFT --output MSFT_DCF.xlsx
```

Option B — uploaded historicals. Prepare a CSV or Excel file with one row per fiscal year and a Revenue column. Add as many of the other line items as you have. See sample_historical_financials.csv.

```
python generic_dcf_from_scratch.py --file sample_historical_financials.csv --output MyCompany_DCF.xlsx
```

4. Answer the assumption prompts

The program displays one question at a time. Press Enter to accept the suggested default, or type a value. For percentages, type either 8% or 0.08. Use the same financial units throughout the model. Do not type a market price in rupees if the equity value and share count imply a different unit scale.

You will be asked for forecast years, revenue growth, EBITDA margin, D&A;/revenue, CapEx/revenue, working-capital assumption, tax rate, risk-free rate, beta, equity risk premium, cost of debt, debt weight, terminal growth, cash, debt, shares outstanding and current share price.

5. Open and review the workbook

After the program finishes, open the output XLSX in Excel. If Excel asks whether to update or recalculate formulas, allow recalculation. Start with the Financials sheet: verify every year, line item, unit and missing value. Then review the yellow input cells on Assumptions and confirm that the assumptions are appropriate.

6. Understand the five sheets

Financials: historical values and notes. **Assumptions:** editable assumptions in pale yellow; calculated WACC-related values in pale green. **Forecast:** projected income statement and UFCF. **DCF & Sensitivity:** enterprise value, equity value, implied share value, upside/downside and a WACC/terminal-growth grid. **Charts & Dashboard:** KPI cards and two charts.

7. Important checks

• Terminal growth must be lower than WACC. • Check cash and debt are in the same units as enterprise value. • Check share count units against the share price currency/unit. • Confirm that yfinance did not omit an important line item. • Review negative CapEx or working-capital signs. • Do not treat default assumptions as recommendations.

8. Troubleshooting

No yfinance data: use the uploaded CSV/XLSX mode. **Missing financial items:** review the Financials sheet and update the source data; do not assume blank data means zero. **Excel shows formulas instead of values:** set calculation to Automatic and recalculate. **Valuation seems implausible:** verify units, shares outstanding, debt/cash, WACC, terminal growth and assumptions.

9. Files included

generic_dcf_from_scratch.py — main program; requirements_generic_dcf.txt — dependencies; sample_historical_financials.csv — example upload format; Sample_Generic_DCF_Model.xlsx — example workbook generated using sample data; MASTER_PROMPT_Generic_DCF_From_Scratch.txt — reusable prompt for another AI to rebuild or extend the project; the technical documentation PDF explains the design and formulas.

Disclaimer

This tool is for learning and analytical support. It is not investment advice. Validate source data and assumptions before making decisions.